

Fotozen AI — Debug & Release Build Guide

Direct Canon USB (Android TV Mini PC)

Use these builds so **direct Canon USB** is the default. The app uses Canon **EDSDK** on the Mini PC to show Pose live preview and fire the shutter in software — no Raspberry Pi and no HDMI capture card.

Run Flutter commands from `photobooth/` unless noted.

0. One-time / after C++ sidecar changes

Only needed if you changed `canon_sidecar/` or JNI spawn:

```
cd /Users/janakiram/photobooth/canon_sidecar
SIDECAR_ARCHES=arm32 ./build.sh
```

Then a **full** Flutter install (not hot reload).

1. GSM (required so backend does not send the app to a Pi)

On this kiosk in ZenAI / GSM:

- `cameraConnectionMode = direct`
- Leave Pi host unused (or ignore it; `direct` overrides it)

Without that, an old Pi IP can still be inferred if the mode field is empty.

2. Camera settings for Direct Canon EDSDK

The sidecar talks to the camera over **USB PTP**. It then programmatically:

- Turns on live view and routes EVF to the PC (`Evf_Mode` + `Evf_OutputDevice = PC`)
- Polls JPEG frames for Pose (`POST /camera/preview`)
- Fires the shutter and downloads a JPEG (`POST /camera/capture`)

You do **not** start live view or press the shutter by hand. You only put the body in a USB-PC state the SDK can control.

2a. Menu and dial (do this before plugging USB)

Setting	Set to	Why
USB connection	PTP (or PC / PC connection)	EDSDK needs PTP. Not MTP, Print/MTP, or Mass storage.
Auto power off	Disable	Sleep drops the USB session (<code>connected: false</code>).
Mode dial	M (Manual stills)	Most reliable for programmatic AF + shutter. Not movie / video.
Shooting mode	Stills (photo)	Video mode blocks still capture.

Lens cap	Off	AF and exposure fail with the cap on.
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USB menu path (typical EOS / Rebel, including 200D II / SL3):

1. Press **Menu**.
2. Open the **wrench / Set-up** tab.
3. **Communication settings** (sometimes under **Wi-Fi / Bluetooth**).
4. **USB connection** → **PTP**.

If the body has no USB connection item, it usually defaults to PTP when plugged into a PC — leave it.

Other labels you may see: **PC connection**, **PC**, **PTP/MTP** → choose **PTP** or **PC**.

2b. Cable: USB to the Mini PC (not a Pi)

1. Use the **Canon USB cable** (camera Mini-B or Micro-B → USB-A).
2. Plug USB-A into a **USB port on the Android TV Mini PC** (the box running Fotozen).
3. Do **not** plug that cable into a Raspberry Pi, a hub that only goes to the Pi, or a laptop running EOS Utility (only one host can own PTP).
4. **HDMI is not required** for this path. Pose preview is USB EVF over EDSDK, not a capture card.

2c. Android USB permission (once per camera / reboot)

When the camera is attached, the TV should show:

```
Allow Fotozen AI to access Canon ...?
[ ] Always open Fotozen AI when this device is connected
[Cancel]    [OK]
```

Select **OK**. Tick **Always open** if you want to skip the dialog after reboot.

If no dialog appears: unplug/replug USB, or:

```
adb shell am start -n com.srisarani.fotozenai/.MainActivity
```

2d. Recommended (kiosk reliability)

- Full battery, or dummy battery / AC coupler.
- AF: One-Shot is fine in decent light; in a dark booth **M** + less hunting is more reliable (sidecar tries Halfway → Completely AF, then Non-AF shutter).
- Image quality on the body can stay JPEG; the sidecar requests **Large JPEG Fine** for download.
- Do not open the camera in playback-only with USB busy on another computer.

2e. Do not use these camera modes

Avoid	What happens
MTP / Mass storage	Sidecar cannot open an EDSDK session (<code>device busy</code> / no camera).
Movie / video dial	Live view or capture times out.
Auto power off still on	Preview dies after a few minutes.
USB to Pi and Mini PC	Wrong host; Pose talks to localhost EDSDK on the TV box.

3. Debug (day-to-day)

Impeller **must** be off on this box. Sidecar dart-defines already default to localhost USB;

`CAMERA_CONNECTION_MODE=direct` makes that explicit even if GSM mode is blank.

Run on the box

```
cd /Users/janakiram/photobooth/photobooth
flutter pub get
flutter run --debug --no-enable-impeller \
  --dart-define=CAMERA_CONNECTION_MODE=direct
```

Build a debug APK

```
cd /Users/janakiram/photobooth/photobooth
flutter pub get
./scripts/flutter_with_version.sh build apk --debug --no-enable-impeller \
  --dart-define=CAMERA_CONNECTION_MODE=direct
```

Output

- `photobooth/build/app/outputs/flutter-apk/app-debug.apk`
- or `photobooth-debug.apk` depending on Gradle naming

Install

```
adb install -r photobooth/build/app/outputs/flutter-apk/photobooth-debug.apk
# if that path is missing:
adb install -r photobooth/build/app/outputs/flutter-apk/app-debug.apk
```

After install

```
adb shell am force-stop com.srisarani.fotozenai
adb shell am start -n com.srisarani.fotozenai/.MainActivity
```

4. Release (kiosk APK)

Do **not** use plain `flutter build apk`. Release must sync version + Bugsnag key, and **must** disable Impeller or Pose can be blank/frozen.

Put `BUGSNAG_API_KEY` in `photobooth/.env` (copy from `.env.example` if needed).

```
cd /Users/janakiram/photobooth/photobooth
flutter pub get
./scripts/flutter_with_version.sh build apk --release --no-enable-impeller \
  --dart-define=CAMERA_CONNECTION_MODE=direct
```

Output

- `photobooth/build/app/outputs/flutter-apk/photobooth-release.apk`
- or `app-release.apk`

Install (uninstall debug first if signatures conflict):

```
adb uninstall com.srisarani.fotozenai
adb install -r photobooth/build/app/outputs/flutter-apk/photobooth-release.apk
adb shell am start -n com.srisarani.fotozenai/.MainActivity
```

Optional dart-defines (already the code defaults; only needed to override):

- CAMERA_SIDE CAR_ENABLED=true
- CAMERA_SIDE CAR_URL=http://127.0.0.1:8791
- CAMERA_SIDE CAR_LIVE_PREVIEW=true

5. Confirm preview and programmatic snap

1. Camera settings from **section 2** applied; USB into the Mini PC.
2. USB permission granted.
3. Open Pose — you should see Canon EVF (software live view), not HDMI/UVC / “Starting camera...”.
4. Take a photo in the app — the body should fire by itself (EDSDK shutter), then preview should resume.

USB enumerated:

```
adb shell lsusb
```

Look for Canon vendor **04a9** , for example ID 04a9:32cb Canon, Inc. EOS 200D II .

Sidecar healthy (app running):

```
adb shell curl http://127.0.0.1:8791/health
```

Expect "ok":true,"connected":true .

Logs:

```
adb logcat -s CanonSidecar:I CanonUsbPerm:I flutter:I
```

Look for camera ready , live view armed, and POSE using Canon USB EVF live preview .

EDSDK error **0x00000021** = device busy (wrong USB mode or another PC holds the session).

0x00000083 = session already open (force-stop the app and retry).

Do not

Command	Why
flutter build apk (no wrapper)	Skips version sync; release also skips Bugsnag key
Release without --no-enable-impeller	Blank/frozen Pose on this Mini PC
Hot reload after C++/JNI changes	Old sidecar stays running

Pi booths: omit `CAMERA_CONNECTION_MODE=direct` and set GSM `cameraConnectionMode=pi` plus sidecar host/port.